

How Many Householders Does State Tracking Need? A Representation-Theoretic Characterisation for One-Layer Linear RNNs, and Where Training Falls Short

Team Ninja Turtles

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Abstract Why do architectures fail at state tracking? Circuit complexity places log-precision Transformers and diagonal state-space models in TC^0 [C-lit-merrill-tc0][C-lit-illusion]. Bounded-width branching programs, by contrast, recognise exactly NC1 [C-lit-barrington]. We study one-layer linear RNNs whose transitions are products of Householder factors, as in DeltaNet and DeltaProduct. We assume real states, finitely many reachable states and token-local transitions, and we allow $\beta = 2$. Under these assumptions, we prove by hand (not machine-checked) that k factors per token suffice if and only if $k \geq h^*(G, \Sigma)$, an invariant defined over covering groups [C-thm1]. We did not find a lower bound on factors per token for general groups and alphabets in prior work; for S_5 alone, Complex KDA proves that four factors are the minimum [C-novelty][C-lit-ckda-four-body]. A_5 with involution inputs needs one reflection. A_5 with all inputs needs two, and so do the S_4/A_5 formats of the closest prior work [C-T-A5-involutions][C-T-A5-all][C-T-S4-tn][C-T-A5-c3c5]. An atlas covers all 319 groups of order at most 63 and determines h^* exactly for 138 of them [C-atlas]. Upper bounds are exact certificates. Lower bounds combine Lean-checked lemmas with a compression lemma proved by hand [C-L1][C-L4][C-L2].

Training falls short of the theory. On the preregistered H-EX1 grid our predictor matched 47 of 61 cells [C-H-accuracy]. The representation baselines matched 50 and 48 [C-H-accuracy]. The H-EX2 criterion was not met [C-X-result]. hh1 did not learn A_5 /involutions, and no architecture learned A_5 /all [C-G-hh1-A5-involutions][C-G-hh4-A5-all]. One cell succeeded in 18 of 20 seeds on the GPU and in 0 of 20 on the CPU [C-H-replication].

1 Introduction

The architecture debate. Transformers with logarithmic precision can be simulated by constant-depth threshold circuits [C-lit-merrill-tc0]. State-space models “cannot express computation outside the complexity class TC^0 ”, so they “cannot solve simple state-tracking problems like permutation composition” [C-lit-illusion]. Bounded-width polynomial-size branching programs recognise exactly the languages in NC1 [C-lit-barrington]. Under standard complexity conjectures, this separates TC^0 architectures from state tracking [C-lit-rwkv7].

Self-attention has further limits on periodic finite-state languages [C-lit-hahn]. In experiments, LSTMs solve regular and counter-language tasks, while RNNs and Transformers fail to generalise on non-regular ones [C-lit-deletang]. Transformers find shallow shortcuts to automata, and Krohn-Rhodes theory helps explain them [C-lit-liu]. Chain of thought increases the expressive power of Transformers [C-lit-cot][C-lit-li-cot].

The gap. Several constructions show that linear RNNs can track state:

- Linear RNNs whose transitions are products of identity-minus-outer-product matrices “can learn any regular language”. At finite precision, with only positive eigenvalues, they cannot solve parity [C-lit-grazzi].
- RWKV-7 recognises all regular languages [C-lit-rwkv7] and can track swaps on five elements [C-lit-rwkv7-

swaps-body].

- DeltaProduct uses products of n_h generalized Householder transformations, and its state tracking improves as n_h grows [C-lit-deltaproduct].

These constructions give upper bounds. DeltaProduct also reports an unexpected result: “ S_4 and A_5 can extrapolate robustly using only $n_h = 2$ despite the theorem suggesting 3 and 4, respectively” [C-lit-deltaproduct-body].

Howe states a representation law: “the minimal n_h that length-generalizes equals the maximal generator reflection length $\text{rank}(I - P)$ in the representation pinned by the task format (parity 1, S_4 3, A_5 / S_5 4)” [C-lit-howe-law-body]. The formats are “ S_4 (transposition + 4-cycle), A_5 (3-cycle + 5-cycle; all-even), S_5 (transposition + 5-cycle; non-solvable)” [C-lit-howe-format-body]. This law is empirical and concerns learning. It is stated for an architecture without the additive input pathway, where input acts only through the orthogonal transitions [C-lit-howe]. Our realisation with a zero initial state uses exactly that additive term [C-L3]. So a gap between our expressivity values and Howe’s learned values is not a contradiction.

Complex KDA (arXiv:2609.24797) proves a lower bound [C-lit-ckda-thm4-body]. It rules out one-layer S_5 tracking for DeltaProduct with at most three factors, assuming non-expansive transitions and finite reachability [C-lit-ckda-thm4-body]. Under those assumptions, four factors are the minimum for S_5 [C-lit-ckda-four-body]. Its compression starts from a word of minimum Frobenius norm [C-lit-

ckda-compression-body], and the paper notes that tracking does not require a faithful representation [C-lit-ckda-decoder-body]. Our compression is different. It uses the minimal ideal of the transition monoid, needs no norm bound, and transfers the rank bound. Neither result contains the other [C-ckda-relation]:

- Complex KDA covers scalar gates, its own parameterisation, and several heads.
- Our result covers expansive transitions and every finite group and alphabet.

We found no general minimum for an arbitrary finite group and input alphabet in prior work [C-novelty][C-ckda-relation].

Contributions.

- *An exact law, proved by hand.* For every finite group G and generating alphabet Σ , a real, finite-state, token-local one-layer realisation (with $\beta = 2$ allowed) with k Householder factors per token exists if and only if $k \geq h^*(G, \Sigma)$ [C-thm1]. Here h^* minimises over covering groups and their faithful real representations. We did not find this invariant in prior work; the closest notion, generation in codimension k , has no alphabet [C-novelty][C-lit-martino-singh-body]. The necessity half rests on a compression lemma that needs no norm bound and is proved by hand [C-L2]. The compression idea appears for S_5 in Complex KDA [C-novelty].
- *Machine-checked rank lemmas* in Lean 4 [C-L1][C-L4].
- *Certified instances.*
 - One reflection suffices for A_5 with involution inputs [C-T-A5-involutions]. Complex KDA does not state this [C-ckda-relation].
 - A_5 with all inputs needs exactly two [C-T-A5-all].
 - The generator formats of prior work need two [C-T-S4-tn][C-T-A5-c3c5]. The upper bound of two comes from DeltaProduct’s $SO(3)$ construction [C-lit-deltaproduct-so3-body][C-lit-deltaproduct-body]. We add the matching lower bound, from L4 and L2.
- S_5 , from a hand step plus a cited theorem, building on Complex KDA. We obtain $h^*(S_5, \text{all}) = 4$, but this is not a certified instance [C-S5][C-T-S5-all].
- *An atlas* over all groups of order at most 63 [C-atlas]. We did not find a comparable atlas in prior work [C-novelty].
- *A preregistered training grid and addendum.* Neither preregistered success criterion was met; on the grid, both representation baselines predicted learnability better than our invariant [C-H-gates][C-X-result][C-H-accuracy].

2 Setting and definitions

Let Σ be an alphabet of letters that generate a finite group G . A word over Σ is read token by token. Solving the word problem means outputting the group element of every prefix, for every length [C-L2][C-thm1].

A *one-layer realisation* is a real recurrence $z_t = A(s_t)z_{t-1} + B(s_t)$ with an arbitrary readout. The transition A and the input term B depend only on the current token s_t [C-L2]. So transitions are token-local, with no short convolution [C-thm1]. The realisation is *finite-state* if only finitely many states are reachable [C-L2][C-thm1].

Transition families.

- A Householder family with k factors per token uses $A(s) = \prod_{i=1}^k (I - \beta_i u_i u_i^T)$ [C-L1]. The models hh1–hh4 are DeltaNet/DeltaProduct with $\beta \in [0, 2]$ [C-H-protocol].
- `diag_pos` has diagonal transitions with entries in $[0, 1]$ [C-L5].
- `diag_pm` has real diagonal entries in $[-1, 1]$ [C-L5].
- `cdiag` has complex diagonal entries of modulus at most 1 [C-L5].

The invariants. $h^*(G, \Sigma)$ is the least k for which all of the following exist [C-thm1]:

- a finite group H ,
- a surjection $\pi : H \rightarrow G$,
- generating lifts t_s of the letters,
- a faithful real representation ρ of H with $\text{rank}(\rho(t_s) - I) \leq k$ for every letter s .

$h(G, \Sigma)$ is the same minimum with $H = G$ [C-thm1]. Requiring the lifts to generate H does not change h^* [C-thm1].

The definitions and all written proofs of the lemmas and of Theorem 1 are in the companion ledger [C-ledger].

3 Results I: theory

Lemma L1 (machine-checked). If $R_1, \dots, R_k$ have rank at most one, then $\text{rank}(\prod_{i=1}^k (I + R_i) - I) \leq k$. In particular, a token transition made of k factors $I - \beta u u^T$ differs from the identity by a matrix of rank at most k [C-L1]. The proof is in Lean 4 with Mathlib and uses only the standard axioms. As a negative control, Lean refutes the strengthened bound $k - 1$ [C-L1].

Lemma L4 (machine-checked). A real matrix of finite order with $\text{rank}(M - I) \leq 1$ is an involution. Hence, if ρ is a faithful real representation of H , $\pi : H \rightarrow G$, and $\pi(t) = s$ with s of order at least 3, then $\text{rank}(\rho(t) - I) > 1$. So $h^*(G, \Sigma) \geq 2$ whenever Σ contains a letter of order at least 3, even when a larger covering group is used [C-L4]. That a real reflection has order two is classical [C-novelty].

Lemma L2 (compression; proved by hand, not machine-checked). Suppose a one-layer real recurrence with arbitrary readout has three properties: it solves the word problem of (G, Σ) for every length, it has finitely many reachable states, and $\text{rank}(A(s) - I) \leq k$ for every letter. Then $h^*(G, \Sigma) \leq k$ [C-L2].

The proof has three steps [C-L2]:

1. The reachable states form a finite transformation monoid that maps onto G .
2. An idempotent e of its minimal ideal gives a group eTe that maps onto G .

3. Restrict the compressed maps $A(u)A(s)$ to the column space of the affine span of $e(Q)$. They form a faithful representation of a covering group and satisfy $\text{rank}(A(u)(A(s) - I)) \leq k$.

The lemma covers affine input terms, $\beta \in [0, 2]$ including singular transitions, matrix-valued states, and arbitrary readouts. Two independent red-team agents reviewed it. The second checked the construction exactly on adversarial instances and found it valid after wording fixes [C-L2].

Lemma L3 (sufficiency; proved by hand, re-verified on every concrete instance by the exact verifier). Suppose H maps onto G with generating lifts t_s and a faithful real representation ρ with $\text{rank}(\rho(t_s) - I) \leq k$. Then an exact finite-state realisation with k reflections per token exists in dimension $\dim \rho$, with each β equal to either 0 or 2 [C-L3]. The construction has four steps:

1. An H -invariant inner product makes ρ orthogonal.
2. Cartan–Dieudonné factors each $\rho(t_s)$ into $\text{rank}(\rho(t_s) - I)$ reflections.
3. A generic initial state separates the elements of H .
4. The readout maps $\rho(x)z_0$ to $\pi(x)$.

The construction needs $\beta = 2$ exactly, with unit keys. With a zero initial state, as in DeltaNet, the input term $v = -ck$ supplies the offset [C-L3]. Prior work uses the classical fact that an $n \times n$ orthogonal matrix is a product of n reflections to obtain upper bounds [C-lit-grazzi-cd-body].

Theorem 1 (proved by hand from Lemmas L2 and L3; proof in the companion ledger [C-ledger]). For every finite group G and generating alphabet Σ , a finite-state one-layer realisation of the word problem with k Householder factors per token exists if and only if $k \geq h^*(G, \Sigma)$. The hypotheses are [C-thm1]:

- real states;
- finitely many reachable states;
- transitions that depend only on the current token;
- one layer;
- β allowed to equal 2.

What is proved how.

- *Machine-checked:* L1 and L4 [C-L1][C-L4].
- *Hand proofs, not machine-checked* [C-L2][C-L3][C-thm1][C-L5][C-L7][C-open-beta][C-complex][C-S5][C-Q8-lower]:
 - Lemmas L2, L3, L5 and L7;
 - Theorem 1;
 - the open- β corollary;
 - the complex-state remark;
 - the S_5 lower-bound step;
 - the Q_8 lower bound.
- *Exact verifier:* it re-checks every concrete upper bound [C-L3][C-verifier].

Computing h . By standard character theory, $\text{codim Fix } \rho(g) = \dim \rho - \frac{1}{|g|} \sum_j \chi_\rho(g^j)$. Codimensions add

over direct sums. A sum of real irreducible representations is faithful if and only if their kernels intersect trivially. So $h(G, \Sigma)$ is a finite optimisation over the real character table [C-L6].

Diagonal families (Lemma L5; proved by hand via the compression lemma, not machine-checked). A finite-state one-layer realisation exists [C-L5]:

- with entries in $[0, 1]$, only for the trivial group;
- with real entries in $[-1, 1]$, exactly for elementary abelian 2-groups;
- with complex entries of modulus at most 1, exactly for abelian groups.

This agrees with prior theorems on diagonal SSMs [C-L5]. For example, single-layer DCD SSMs cannot track any non-abelian group at finite precision [C-lit-shakerinava].

Abelian groups (Lemma L7; proved by hand, instances certified). For a nontrivial abelian group, $h^*(G, \Sigma)$ is 1 if every letter is an involution and 2 otherwise [C-L7]. The upper bound uses a count cover, which tracks each letter’s count modulo its order. Its dimension is the number of involution letters plus twice the number of other letters. This is an upper bound on the dimension, not the minimum [C-L7].

Boundary remarks (proved by hand; found by the theory red team).

- If β is confined to $[0, 2]$, as with a sigmoid parameterisation, no nontrivial group is exactly realisable for any number of factors [C-open-beta]. Likewise, open intervals for diagonal entries admit only the trivial group [C-open-beta].
- A complex realisation of complex rank k only gives $h^* \leq 2k$ [C-complex].

4 Results II: certified instances

Table 1 lists every certified instance. The two bounds in each row have different status:

- *Upper bound:* an exact certificate.
- *Lower bound:* the hand-proved Lemma L2, combined with either Lemma L4 (machine-checked) or the non-triviality of G [C-T-A5-all][C-T-Z3-all][C-T-Z2-all].

A_5 with involution inputs: one reflection. In the $\text{SO}(3)$ construction, every involution letter is lifted to its negative. This gives an exact realisation with one reflection per token, through a covering group of order 120 in dimension 3 [C-T-A5-involutions]. So $h^*(A_5, \text{involutions}) = 1$, although the faithful value is $h = 2$ [C-T-A5-involutions]. By Lemma L4, no such lift exists for letters of order at least three [C-L4].

A_5 with all inputs: two. $h^*(A_5, \text{all}) = 2$, via the $\text{SO}(3)$ construction in dimension 3. The permutation representation costs 4 [C-T-A5-all]. The same value holds for the 3-cycle and 5-cycle alphabets [C-T-A5-cycles3][C-T-A5-cycles5].

Generator formats of prior work.

- $h^*(S_4, \text{tn}) = 2$, via so3 in dimension 3 [C-T-S4-tn].
- $h^*(A_5, \text{c3c5}) = 2$, via so3 [C-T-A5-c3c5].

In the permutation representation these formats cost 3 and 4 [C-T-S4-tn][C-T-A5-c3c5], which are the values of Howe’s law [C-lit-howe-law-body]. Howe’s law is an empirical statement about length generalisation without the additive input pathway [C-lit-howe]. Our values are exact expressivity values for a realisation that may use the additive term [C-L3]. The lower values here therefore do not contradict that law.

The $SO(3)$ upper-bound constructions for S_4 and A_5 are due to DeltaProduct, which attributes their efficiency to “their isomorphism to subgroups of $SO(3, \mathbb{R})$ ” [C-lit-deltaproduct-so3-body]. Complex KDA states a related one-head result for subgroups of $SO(3)$ [C-lit-ckda-so3-body]. For these formats we contribute the matching lower bound [C-T-S4-tn][C-T-A5-c3c5]. Under our hypotheses, two factors are exactly optimal for these formats in exact expressivity [C-T-S4-tn][C-T-A5-c3c5]. This is consistent with DeltaProduct’s unexpected $n_h = 2$ observation, but an expressivity bound alone does not explain a learning result [C-lit-deltaproduct-body].

Transposition alphabets. S_3 , S_4 and S_5 with transposition alphabets each need exactly one reflection [C-T-S3-transpositions][C-T-S4-transpositions][C-T-S5-transpositions].

$\mathbb{Z}_2^3$: **below the faithful value.** $h^*(\mathbb{Z}_2^3, \text{all}) = 1$, via a count cover of order 128 in dimension 7, while the faithful value is $h = 3$ [C-T-Z2p3-all]. Among the diagonal families, `diag_pm` and `cdiag` realise it and `diag_pos` does not [C-T-Z2p3-all].

Q_8 .

- The faithful value is $h = 4$, and the certified lower bound is 2 [C-T-Q8-all].
- The covering group C4:C4 of order 16, with a rational 4-dimensional representation, certifies $h^*(Q_8, \text{all}) \leq 3$ [C-T-Q8-cover].
- A hand argument, not machine-checked, gives $h^*(Q_8, \text{all}) \geq 3$ [C-Q8-lower].

S_5 , **kept separate.** For both alphabets `all` and `tn`, the certified bounds are $[2, 4]$ and the faithful value is $h = 4$ [C-T-S5-all][C-T-S5-tn]. The value $h^*(S_5, \text{all}) = 4$, under exact finite reachability, rests on a hand step combined with a published theorem [C-S5]:

1. A cover with rank at most 3 can be made orthogonal by averaging.
2. It is then a single-head, orthogonal, finite-state tracker of the kind excluded by Theorem 4 of arXiv:2609.24797 [C-S5][C-lit-ckda-thm4-body].

Our argument only needs an alphabet that contains a 5-cycle and a transposition [C-S5]. This extension is our own observation, proved by hand; it is not stated in arXiv:2609.24797 [C-S5].

An independent hand argument gives $h^*(S_5, \text{all}) \geq 3$ and $h^*(S_5, \text{tn}) \geq 3$ unconditionally. Combined with the classification of arXiv:1509.06922, it gives 4 for every alphabet that contains a 5-cycle [C-S5-hand3][C-lit-lange-mikhailova-body]. For S_5 the value 4 matches Howe’s law [C-lit-howe-law-body]. Here the two agree, whereas on S_4 and A_5 they differ.

Contrast with circuit complexity. The non-solvable A_5 with all inputs costs exactly as much as the solvable S_3 [C-T-A5-all][C-T-S3-all], the cyclic $\mathbb{Z}_3$ [C-T-Z3-all], or A_4 [C-T-A4-all]. With involution inputs, A_5 costs as much as parity [C-T-A5-involutions][C-T-Z2-all]. In these instances solvability does not determine the Householder budget; under Theorem 1 (proved by hand) it is set by the alphabet and the representation theory [C-thm1].

5 Results III: the atlas

The atlas covers all 319 groups of order at most 63 in the GAP SmallGroups library, with the full alphabet [C-atlas]. Figure 2 and Table 3 summarise it.

- Our numerical character tables and GAP’s exact tables give the same faithful h for every group (0 disagreements) [C-atlas].
- h^* is determined exactly for 138 of the 319 groups [C-atlas]. These are 32 of the 213 non-abelian groups, plus every abelian group by Lemma L7 [C-atlas].
- Over the non-abelian groups, the faithful h is distributed as 2: 32, 3: 8, 4: 115, 5: 11, 6: 41, 8: 5, 10: 1 [C-atlas-dist].
- Sign lifts to $G \times \mathbb{Z}_2$ push the certified upper bound below h for 37 of the 213 non-abelian groups [C-atlas-twist].
- Covers of twice the group order lowered 17 further upper bounds [C-atlas-improved]. Their lower bounds stay at 2, so these groups remain undetermined [C-atlas-improved].

Non-monotonicity. The only non-solvable group in the atlas is A_5 , with $h = h^* = 2$ [C-atlas-nonsolvable]. 181 of the 212 solvable non-abelian groups have faithful $h \geq 3$ [C-atlas-monotone]. The largest faithful value is 10 [C-atlas-monotone]. So, in faithful representations, many solvable groups need more Householder factors per token than A_5 . For h^* , however, these groups only have certified bounds $[2, \text{upper}]$, so we claim no strict separation in h^* [C-atlas-monotone].

6 Results IV: preregistered experiments

6.1 H-EX1 grid

Protocol. The protocol was committed before the first confirmatory run [C-H-protocol]:

- *Grid:* 9 tasks $\times$ 7 architectures (`diag_pos`, `diag_pm`, `hh1-hh4` with $\beta \in [0, 2]$, and an LSTM) [C-H-protocol].
- *Model:* one layer, 64-dimensional state, MLP readout [C-H-protocol].
- *Training:* 20 seeds (1000-1019), 3000 steps, a length curriculum from 8 to 64 and a final stage at 128 [C-H-protocol].
- *Seed success:* token accuracy of at least 0.9 on positions 257-512 of length-512 sequences [C-H-protocol].

- *Cell success*: at least 10 of 20 seeds succeed [C-H-protocol].

The parameterisation reaches $\beta = 0$ and $\beta = 2$ exactly [C-open-beta]. For every highlighted cell we also report the preregistered secondary outcome at 7x-8x the longest training length [C-G-diag_pm-Z2-all].

Predictors. We compare our algebraic predictor with five baselines [C-H-accuracy]:

- circuit class (solvable);
- abelian;
- group size;
- the representation law in the permutation representation;
- faithful representations only.

Accuracy (primary). Agreement with the outcomes on the determined cells [C-H-accuracy]:

Predictor	Cells matched
Our predictor	47 of 61 [C-H-accuracy]
Circuit class	41 of 61 [C-H-accuracy]
Abelian	37 of 61 [C-H-accuracy]
Group size	41 of 61 [C-H-accuracy]
Permutation-representation law	50 of 61 [C-H-accuracy]
Faithful representations only	48 of 61 [C-H-accuracy]

Both representation baselines matched more cells than our predictor. The preregistered criterion “accuracy beats every baseline” was therefore not met [C-H-gates]. Figure 1 and Table 2 show all cells.

Every cell where our predictor errs. Counts are successful seeds at 2x-4x the longest training length. The secondary outcome at 7x-8x is given in parentheses [C-G-diag_pm-Z2-all]. In all fourteen cells, our predictor predicted success and the cell failed:

- hh2 on $\mathbb{Z}_2$ /all: 3 of 20 (9 of 20) [C-G-hh2-Z2-all]
- hh4 on $\mathbb{Z}_2$ /all: 6 of 20 (7 of 20) [C-G-hh4-Z2-all]
- hh3 on $\mathbb{Z}_3$ /all: 6 of 20 (5 of 20) [C-G-hh3-Z3-all]
- hh1 on $\mathbb{Z}_2^3$ /all: 1 of 20 (1 of 20) [C-G-hh1-Z2p3-all]
- hh1 on A_5 /involutions: 0 of 20 (0 of 20) [C-G-hh1-A5-involutions]
- hh2 on A_5 /involutions: 0 of 20 (0 of 20) [C-G-hh2-A5-involutions]
- LSTM on A_5 /involutions: 0 of 20 (0 of 20) [C-G-lstm-A5-involutions]
- hh2 on A_5 /all: 0 of 20 (0 of 20) [C-G-hh2-A5-all]
- hh3 on A_5 /all: 0 of 20 (0 of 20) [C-G-hh3-A5-all]
- hh4 on A_5 /all: 0 of 20 (0 of 20) [C-G-hh4-A5-all]
- LSTM on A_5 /all: 0 of 20 (0 of 20) [C-G-lstm-A5-all]
- LSTM on S_5 /transpositions: 8 of 20 (1 of 20) [C-G-lstm-S5-transpositions]
- hh4 on S_5 /all: 0 of 20 (0 of 20) [C-G-hh4-S5-all]
- LSTM on S_5 /all: 0 of 20 (0 of 20) [C-G-lstm-S5-all]

For hh2 and hh3 on S_5 /all our predictor is undetermined. Both cells had 0 of 20 [C-G-hh2-S5-all][C-G-hh3-S5-all].

hh1 did not learn the headline one-reflection instance, A_5 /involutions. Its in-distribution accuracy there was only 0.074 [C-G-hh1-A5-involutions]. No architecture learned

A_5 /all [C-G-diag_pos-A5-all][C-G-diag_pm-A5-all][C-G-hh1-A5-all][C-G-hh2-A5-all][C-G-hh3-A5-all][C-G-hh4-A5-all][C-G-lstm-A5-all].

Highlighted cells where the prediction matched.

- hh1 on S_3 /transpositions: 20 of 20 (18 of 20) [C-G-hh1-S3-transpositions].
- hh1 on S_5 /transpositions: 20 of 20 (20 of 20) [C-G-hh1-S5-transpositions].
- hh1 on S_3 /all fails as predicted: 0 of 20 (0 of 20) [C-G-hh1-S3-all].
- hh1 on $\mathbb{Z}_3$ /all fails as predicted: 0 of 20 (0 of 20) [C-G-hh1-Z3-all].
- hh3 on A_5 /involutions: 20 of 20 (3 of 20) [C-G-hh3-A5-involutions].
- hh4 on A_5 /involutions: 20 of 20 (0 of 20) [C-G-hh4-A5-involutions].
- diag_pm on $\mathbb{Z}_2^3$ /all: 20 of 20 (0 of 20) [C-G-diag_pm-Z2p3-all].
- diag_pm on $\mathbb{Z}_3$ /all fails as predicted: 0 of 20 (0 of 20) [C-G-diag_pm-Z3-all].

Many of these primary successes collapse at 7x-8x [C-G-diag_pm-Z2p3-all]:

- diag_pm on $\mathbb{Z}_2^3$ /all falls from 20 to 0 [C-G-diag_pm-Z2p3-all].
- hh3 on $\mathbb{Z}_2^3$ /all falls from 20 to 0 [C-G-hh3-Z2p3-all].
- hh4 on A_5 /involutions falls from 20 to 0, and hh3 keeps only 3 [C-G-hh4-A5-involutions][C-G-hh3-A5-involutions].
- hh2 on S_3 /all falls from 19 to 1 [C-G-hh2-S3-all].

Agreement on the primary outcome therefore does not imply robust length generalisation.

Discriminating tests. All tests are one-sided Fisher exact tests with Benjamini–Hochberg correction ($m = 9$, $q = 0.1$) [C-H-D1]. We also report BH over all 12 tests of H-EX1 and H-EX2 [C-H-D1].

- **D1:** hh1 on A_5 /involutions vs A_5 /all, testing one reflection for A_5 /involutions. 0 vs 0 successes, $p = 1.000$, not significant [C-H-D1]. The supplementary paired permutation test gave $p = 0.145$ (bootstrap CI 0.99-1.03), so the project criterion was not met [C-H-D1].
- **D2:** hh2 vs hh1 on A_5 /all, testing two factors for A_5 /all. 0 vs 0, $p = 1.000$, not significant [C-H-D2]. Supplementary $p = 0.946$; criterion not met [C-H-D2].
- **D3:** hh1 on $\mathbb{Z}_2^3$ vs $\mathbb{Z}_3$. 1 vs 0, adjusted $p = 1.000$, not significant; BH over all 12 tests gives 0.857 [C-H-D3]. The supplementary paired permutation test ($p = 5.0e-05$, accuracy ratio 1.98) met the project criterion [C-H-D3].
- **D4:** hh1 on S_5 /transpositions vs S_5 /all. 20 vs 0, significant [C-H-D4].
- **D5:** hh1 on $\mathbb{Z}_2$ vs $\mathbb{Z}_3$. 18 vs 0, significant [C-H-D5].
- **D6:** diag_pm on $\mathbb{Z}_2^3$ vs $\mathbb{Z}_3$. 20 vs 0, significant [C-H-D6].

The tests that separate our predictor from the representation baselines are D1–D3, and none of them was significant [C-H-D1][C-H-D2][C-H-D3]. D4–D6 do not separate them:

- In D4 the permutation representation costs 1 against 4, and in D5 1 against 2. This is the same split as h^* [C-T-S5-transpositions][C-T-S5-all][C-T-Z2-all][C-T-Z3-all].
- D6 tests a diagonal family. Its prediction follows from Lemma L5 and agrees with prior theorems on diagonal SSMs [C-L5].

D5 needs a further caveat. Its 18 successes come from the MPS run, for which only the success count survives, so no paired test is possible [C-G-hh1-Z2-all]. The CPU replication of the same cell gave 0 of 20 [C-H-replication]. With the CPU run, D5 would compare 0 with 0 successes [C-H-replication]. H-EX1 therefore gives no evidence for our predictor over the representation baselines. The criterion “all six discriminating tests significant” was not met [C-H-gates].

Controls. The three random-target negative controls (hh4 and LSTM on S_5 , hh2 on A_5) each had 0 successes [C-H-NC-hh4-S5][C-H-NC-lstm-S5][C-H-NC-hh2-A5]. The negative-control criterion held, but H-EX1 as a whole failed [C-H-gates]. The LSTM positive control failed on the A_5 and S_5 tasks [C-H-exploratory].

Exploratory analysis (post hoc, not preregistered). We restricted the comparison to the 5 tasks where the LSTM succeeded [C-H-exploratory]. There, our predictor, the permutation law and the faithful-only predictor each matched 26 of 30 cells [C-H-exploratory]. The other baselines matched 16 or fewer [C-H-exploratory]. On the remaining tasks, the cells mainly measure the training budget [C-H-exploratory].

Device sensitivity. Because of a runner bug, the same protocol and seeds were run once on the MPS GPU and once on the CPU. For hh1 on $\mathbb{Z}_2$ /all, 18 of 20 seeds succeeded on MPS and 0 of 20 on CPU [C-H-replication]. Training outcomes can depend strongly on floating-point details of the device [C-H-replication].

6.2 H-EX2 addendum: the generator formats of prior work

Protocol. The addendum was preregistered before its runs [C-X-protocol]:

- *Tasks:* S_4 /tn and A_5 /c3c5 [C-X-protocol].
- *Architectures:* hh1, hh2 and hh3, with 20 seeds each [C-X-protocol].
- *Setup:* the H-EX1 protocol, run on the MPS GPU [C-X-protocol].
- *Predictions:* we predicted that hh1 fails while hh2 and hh3 succeed. The permutation law predicts that S_4 /tn needs 3 factors and A_5 /c3c5 needs 4 [C-X-protocol].

There is a device confound. H-EX2 ran entirely on MPS; in H-EX1 the hh2–hh4 and LSTM cells ran on the CPU while the primary hh1 and diagonal cells ran on MPS, so device and architecture are partly confounded, and the device changed outcomes in the replication [C-X-protocol][C-H-replication][C-G-hh1-Z3-all]. Table 4 and Figure 3 show the results.

Outcomes. Successful seeds, with the 7x-8x outcome in parentheses [C-X-hh2-A5-c3c5]:

- hh1 on A_5 /c3c5: 0 of 20 (0 of 20) [C-X-hh1-A5-c3c5].
- hh2 on A_5 /c3c5: 18 of 20 (18 of 20) [C-X-hh2-A5-c3c5].
- hh3 on A_5 /c3c5: 18 of 20 (10 of 20) [C-X-hh3-A5-c3c5].
- hh1 on S_4 /tn: 0 of 20 (0 of 20) [C-X-hh1-S4-tn].
- hh2 on S_4 /tn: 6 of 20 (4 of 20) [C-X-hh2-S4-tn]. This is a failure, contrary to our prediction [C-X-hh2-S4-tn].
- hh3 on S_4 /tn: 19 of 20 (3 of 20) [C-X-hh3-S4-tn].

E1 and E3 (on A_5) and E2 (on S_4) are significant after correction [C-X-E1][C-X-E2][C-X-E3]. For E2 the supplementary paired permutation test gives $p = 0.110$ with an accuracy-ratio interval 0.75-2.39 that includes 1, so it does not meet the project criterion [C-X-E2]. Our predictor matched 5 of 6 cells and the permutation law 4 of 6 [C-X-result]. The preregistered success criterion of H-EX2 was not met [C-X-result].

What H-EX2 supports and what it does not.

- *Supported:* under our protocol and readout, two factors learn A_5 /c3c5 (18 of 20 seeds, also 18 of 20 at 7x-8x), below the permutation law’s value of 4 [C-X-hh2-A5-c3c5][C-lit-howe-law-body]. Howe measures length generalisation at 16x the training length, beyond our longest test [C-lit-howe]. This agrees with DeltaProduct’s observation [C-lit-deltaproduct-body]. Howe’s law was stated without the additive input pathway [C-lit-howe], so this is a test under a different architecture variant, not a direct refutation.
- *Not supported:* that two factors are learnable for S_4 /tn, where hh2 failed [C-X-hh2-S4-tn].
- *Scope:* our predictor is a sufficiency statement about exact expressivity [C-thm1]. H-EX2 tests it against a necessity claim of prior work, under one protocol, one device and our readout [C-X-protocol].

7 The agentic lab and the verification pipeline

Agents. The lab had six components [C-lab]:

- a literature scout, whose quotes were checked by code;
- an integrator;
- a cascade of four researcher agents;
- an exact verifier;
- a red-team agent;
- a preregistration for each round.

Runs and outcomes. The lab ran 15 rounds at a cost of 10.51 USD [C-lab].

- 9 rounds produced an accepted claim [C-lab]. All 9 survive re-checking with the hardened verifier, and 0 were withdrawn [C-lab].
- 6 rounds produced a negative result: no claim passed the verifier and the consistency rule [C-lab][C-neg1][C-neg2][C-neg3][C-neg4][C-neg5][C-neg6].

The rounds on Q_8 , A_5 with involutions, and $\mathbb{Z}_2^2$ carry the status “contested by red-team counter-check” [C-expressivity-R4][C-expressivity-R6][C-expressivity-R8].

“Contested” means the red team disputed the round in a counter-check. “Withdrawn” would mean the verified sentence failed re-checking, which happened for no round [C-lab]. The certified $h^*(A_5, \text{all}) = 2$ from the fifth round rests on the hand-proved L2 for its lower bound [C-expressivity-R5].

The agents’ own free-text interpretations were not checked by the verifier and are not results. Several are inconsistent with the hardened consistency rule. For example, one interpretation reads “ > 1 ” for A_5 with involutions [C-expressivity-R3-I], which contradicts the certified value 1 [C-T-A5-involutions]. The scout kept 60 literature findings whose verbatim quote was found by code in the abstract [C-lab].

Loophole found and closed. In round one, an agent answered that one reflection is not enough for A_5 with involution inputs. It supported this with two true but weaker checks, while the verifier had itself found a certificate with $k = 1$ [C-loop-hole1]. The consistency rule now requires an impossibility certificate for negative answers and exact-value checks for numbers [C-loop-hole1].

Verifier. The verifier passed a mandatory self-test of 46 known true and known false claims [C-verifier]. These include near-boundary cases and a regression case for every red-team bug [C-verifier].

Red team. The red team found three bugs [C-redteam]:

1. The group names S1, D1 and D2 denoted the wrong groups.
2. The published h^* sentence omitted the finite-state hypothesis.
3. The consistency rule accepted unchecked numbers in answer texts.

All three were fixed and became regression cases [C-redteam]. The red team also re-checked the exact core [C-redteam]:

- 189 library constructions [C-redteam];
- 15556 exhaustive explicit-matrix claims, against an independent re-implementation [C-redteam];
- the character tables of all 319 SmallGroups, against GAP [C-redteam].

All of these checks gave 0 disagreements [C-redteam]. The theory red team found the open- β corollary and the complex-state remark [C-open-beta][C-complex].

8 Limitations and open questions

- **Hypotheses of the law.** The law assumes finitely many reachable states, token-local transitions without short convolution, and one layer [C-thm1]. Its necessity half rests on a hand proof [C-L2].
- **Real states.** A complex realisation only gives $h^* \leq 2k$ [C-complex].
- **Multi-layer models.** For diagonal SSMs, the number of layers needed is tied to subnormal series [C-lit-shakerinava]. A multi-layer analogue of h^* for Householder products is an open question.

- **Exact arithmetic versus floating point.** The theory is exact [C-L3], while prior analyses work at finite precision [C-lit-grazzi][C-lit-deltaproduct]. Our training outcomes are device-sensitive [C-H-replication].
- **Open parameterisations.** With $\beta \in [0, 2)$, or with open intervals for diagonal entries, only approximation is possible [C-open-beta].
- **Remaining open values.**
 - Q_8 : certified bounds $[2, 3]$; the lower bound 3 is only a hand argument [C-T-Q8-all][C-T-Q8-cover][C-Q8-lower].
 - S_5 : the value 4 rests on hand steps [C-S5][C-S5-hand3].
 - Most non-abelian groups in the atlas remain undetermined [C-atlas].
- **Chain of thought and padding.** Prior work shows that intermediate generation increases expressive power [C-lit-cot][C-lit-li-cot]. Whether intermediate tokens act as an additional Householder budget per input symbol is an open question that we have not tested.
- **Learnability.** The theory predicts exact expressivity, not learnability. On our grid, both representation base-lines predicted training outcomes better than our predictor (50 and 48 vs 47 of 61 cells) [C-H-accuracy][C-H-gates].

Seeds (of 20) that length-generalise; + / - / ? = our predicted outcome

Z2 / all	0-	18+	18+	3+	20+	6+	20+
Z3 / all	0-	0-	0-	10+	6+	12+	19+
Z2^3 / all	0-	20+	1+	20+	20+	20+	20+
S3 / transpositions	0-	0-	20+	10+	19+	18+	20+
S3 / all	0-	0-	0-	19+	20+	19+	20+
A5 / involutions	0-	0-	0+	0+	20+	20+	0+
A5 / all	0-	0-	0-	0+	0+	0+	0+
S5 / transpositions	0-	0-	20+	19+	20+	19+	8+
S5 / all	0-	0-	0-	0?	0?	0+	0+
	diag_pos	diag_pm	hh1	hh2	hh3	hh4	lstm

Figure 1: Preregistered grid: number of seeds (out of 20) whose accuracy on positions 257-512 is at least 0.9, with the outcome predicted by the algebraic law (+ success, - failure, ? undetermined). Table 2 is the table view. Evidence: one claim per cell, C-G-(model)-(task), listed in the claim-trace appendix.

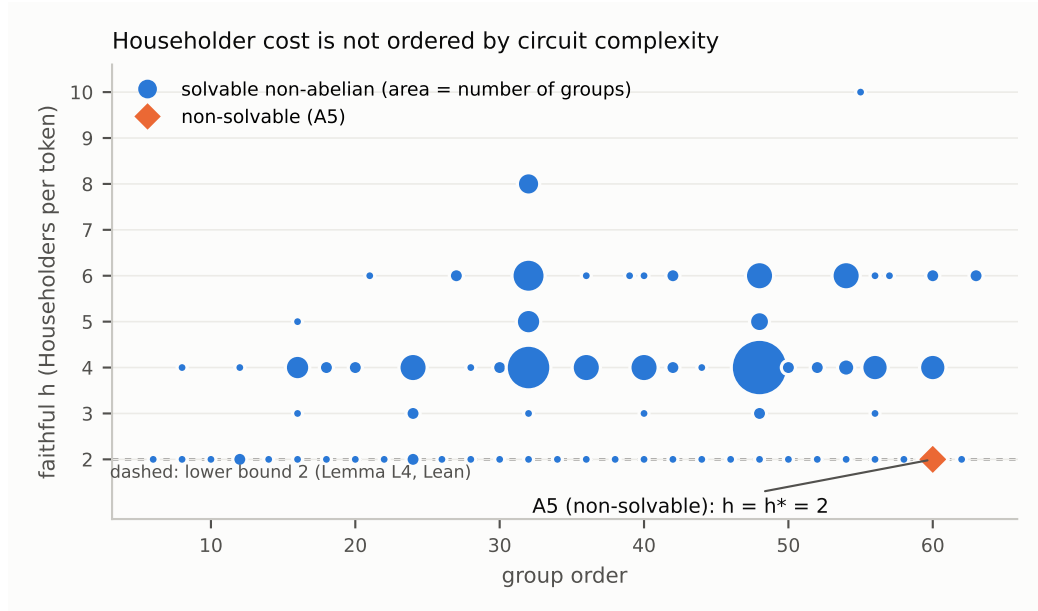


Figure 2: Atlas of all non-abelian groups of order at most 63: faithful h against group order (dot area = number of groups with that order and h). The only non-solvable group, A5, sits at the minimum value 2 while most solvable groups need more. Claims [C-atlas-monotone], [C-atlas-nonsolvable].

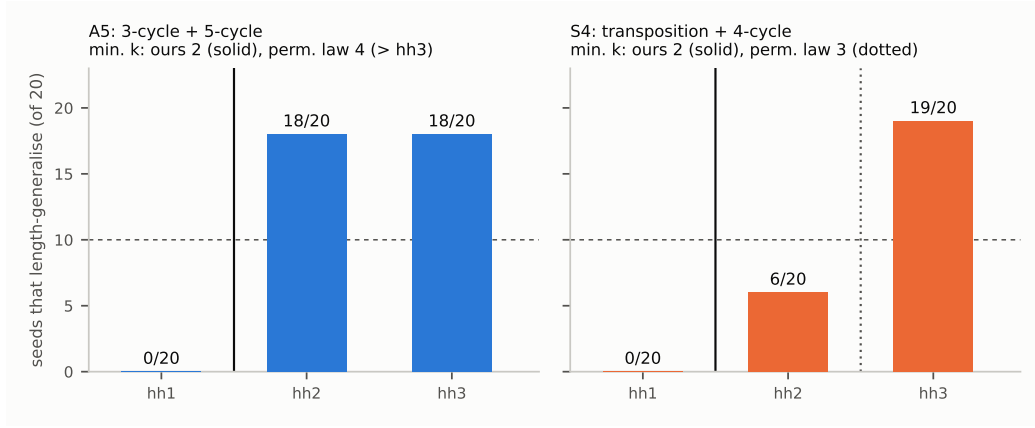


Figure 3: Preregistered addendum H-EX2 on the generator formats of the closest prior work: successful seeds (of 20) for one, two and three Householder factors per token; solid line: minimum predicted by our law ($h^* = 2$, certified); dotted line: the permutation-representation law (3 for S4, 4 for A5); dashed line: the cell-success threshold of 10 seeds. Evidence: claims C-X-(model)-(task) and C-X-E1 to C-X-E3, listed in the claim-trace appendix.

Table 1: Certified one-layer Householder complexity (exact verifier). lower: certified lower bound (Lemma L4 for 2); upper: best exact realisation found by the verifier (construction / sign twist, covering-group order |H|, dimension d); h: faithful representations of G only (own and GAP character tables); perm: cost in the permutation representation (the representation law of prior work). Evidence: claims C-T-(group)-(alphabet) and C-T-Q8-cover, listed in the claim-trace appendix.

G	alphabet	order	solvable	lower	upper	construction	H order	d	h	perm
Z2	all	2	yes	1	1	perm/none	2	2	1	1
Z3	all	3	yes	2	2	perm/none	3	3	2	2
Z5	all	5	yes	2	2	planar/none	5	2	2	4
Z6	all	6	yes	2	2	planar/none	6	2	2	5
Z2^2	all	4	yes	1	1	count/none	8	3	2	2
Z2^3	all	8	yes	1	1	count/none	128	7	3	3
S3	transpositions	6	yes	1	1	perm/none	6	3	1	1
S3	all	6	yes	2	2	perm/none	6	3	2	2
D4	all	8	yes	2	2	planar/none	8	2	2	3
D5	all	10	yes	2	2	planar/none	10	2	2	4
Q8	all	8	yes	2	3	cover C4:C4 (explorer)	16	4	4	6
A4	all	12	yes	2	2	perm/none	12	4	2	2
S4	transpositions	24	yes	1	1	perm/none	24	4	1	1
S4	all	24	yes	2	2	so3/none	24	3	2	3
S4	tn	24	yes	2	2	so3/none	24	3	2	3
A5	c3c5	60	no	2	2	so3/none	60	3	2	4
S5	tn	120	no	2	4	perm/none	120	5	4	4
A5	involutions	60	no	1	1	so3/involutions	120	3	2	2
A5	cycles3	60	no	2	2	perm/none	60	5	2	2
A5	cycles5	60	no	2	2	so3/none	60	3	2	4
A5	all	60	no	2	2	so3/none	60	3	2	4
S5	transpositions	120	no	1	1	perm/none	120	5	1	1
S5	all	120	no	2	4	perm/none	120	5	4	4

Table 2: Preregistered grid: successful seeds out of 20 (accuracy ≥ 0.9 on positions 257-512 of length-512 sequences); mark: predicted success (+), predicted failure (-), undetermined (?) by our predictor. Evidence: claims C-G-(model)-(task).

task	diag_pos	diag_pm	hh1	hh2	hh3	hh4	lstm
Z2/all	0 -	18 +	18 +	3 +	20 +	6 +	20 +
Z3/all	0 -	0 -	0 -	10 +	6 +	12 +	19 +
Z2^3/all	0 -	20 +	1 +	20 +	20 +	20 +	20 +
S3/transpositions	0 -	0 -	20 +	10 +	19 +	18 +	20 +
S3/all	0 -	0 -	0 -	19 +	20 +	19 +	20 +
A5/involutions	0 -	0 -	0 +	0 +	20 +	20 +	0 +
A5/all	0 -	0 -	0 -	0 +	0 +	0 +	0 +
S5/transpositions	0 -	0 -	20 +	19 +	20 +	19 +	8 +
S5/all	0 -	0 -	0 -	0 ?	0 ?	0 +	0 +

Table 3: Atlas, all 319 groups of order ≤ 63 , full alphabet. Evidence: claims C-atlas, C-atlas-nonsolvable, C-atlas-monotone, C-atlas-dist.

class	groups	h^* exact	faithful $h = 1$	faithful $h = 2$	faithful $h \geq 3$
trivial	1	1	0	0	0
abelian (nontrivial)	105	105	1	62	42
solvable non-abelian	212	31	0	31	181
non-solvable	1	1	0	1	0

Table 4: Preregistered addendum H-EX2: the generator formats of the closest prior work. Successful seeds of 20 at 2x-4x (7x-8x); predicted outcome by our law ($h^* = 2$ for both tasks) and by the permutation-representation law. Evidence: claims C-X-(model)-(task), C-X-E1 to C-X-E3.

task	model	seeds 2x-4x	seeds 7x-8x	ours	permutation law
A5/c3c5	hh1	0	0	failure	failure
A5/c3c5	hh2	18	18	success	failure
A5/c3c5	hh3	18	10	success	failure
S4/tn	hh1	0	0	failure	failure
S4/tn	hh2	6	4	success	failure
S4/tn	hh3	19	3	success	success

A Claim trace (Appendix C)

Every statement in the text cites claim identifiers in square brackets. The table lists every claim with its evidence level (proved_lean: Lean 4 + Mathlib; computed_rigorous: exact verifier; statistical: preregistered tests; observed; hypothesis: hand proof or uninspected interpretation) and the start of its canonical text. Full texts: projects/expressivity/paper_evidence.json; proofs of the hand-proved lemmas: expressivity/theory.md (companion ledger).

claim	level	status	canonical text (start)
C-L1	proved_lean	confirmed	Lemma L1 (machine-checked in Lean 4 with Mathlib, theorems rank_prod_sub_one_le and deltaproduct_rank_le): if $R_1, \dots, R_k$ have rank at most one, the...
C-L4	proved_lean	confirmed	Lemma L4 (machine-checked in Lean 4 with Mathlib, theorems sq_eq_one_of_rank_le_one_of_pow_eq_one and no_single_householder_lift): a real matrix of fi...
C-L2	hypothesis	open	Lemma L2 (compression lemma; proved by hand, not machine-checked; reviewed by two independent red-team agents, the second of which checked the constru...
C-L3	hypothesis	open	Lemma L3 (sufficiency; proved by hand, and re-verified on every concrete instance by the exact verifier): if a finite group H maps onto G with generat...
C-thm1	hypothesis	open	Theorem 1 (from Lemma L2, proved by hand, and Lemma L3): for every finite group G and generating alphabet Σ , a finite-state one-layer realisation ...
C-open-beta	hypothesis	open	Corollary (proved by hand; found by the theory red team): if β is confined to $[0, 2)$, as with a sigmoid parameterisation, no nontrivial group is ex...
C-complex	hypothesis	open	Remark (theory red team): the law is stated for real states; a complex realisation of complex rank k only gives $h^* \leq 2k$ (a 1×1 complex rotation track...
C-ledger	observed	confirmed	The definitions and the complete written proofs of Lemmas L2, L3, L4, L5, L7 and Theorem 1 are given in the companion ledger (projects/expressivity/th...

C-L5	hypothesis	open	Lemma L5 (proved by hand via the compression lemma, not machine-checked; consistent with prior theorems on diagonal SSMs): a finite-state one-layer re...
C-L7	hypothesis	open	Lemma L7 (proved by hand; instances certified): for a nontrivial abelian group $h^*(G, \Sigma)$ is 1 if every letter is an involution and 2 otherwise; the...
C-S5	hypothesis	open	Result (lower bound by a hand step combined with a published theorem; upper bound certified): $h^*(S5, \text{all}) = 4$ under exact finite reachability. A cover...
C-ckda-relation	observed	confirmed	Relation to arXiv:2609.24797 (Complex KDA), from a full-text comparison (expressivity/ckda_comparison.md): its compression passes from finite reachabi...
C-novelty	observed	confirmed	Novelty check against the literature (expressivity/novelty.md): the exact law h^* over covering groups with a fixed alphabet and the atlas were not fou...
C-L6	observed	confirmed	Computation of h (standard character theory): $\text{codim Fix } \rho(g) = \dim \rho - (1/g) \sum_j \chi_\rho(g^j)$, codimensions add over direct sums, and a sum of...
C-T-Z2-all	computed_rigorous	confirmed	Certified (Z2, alphabet 'all', order 2, 1 letters, solvable): lower bound 1, best certified realisation $k = 1$ via perm/none (covering group of order 2...
C-T-Z3-all	computed_rigorous	confirmed	Certified (Z3, alphabet 'all', order 3, 2 letters, solvable): lower bound 2, best certified realisation $k = 2$ via perm/none (covering group of order 3...
C-T-Z5-all	computed_rigorous	confirmed	Certified (Z5, alphabet 'all', order 5, 4 letters, solvable): lower bound 2, best certified realisation $k = 2$ via planar/none (covering group of order...
C-T-Z6-all	computed_rigorous	confirmed	Certified (Z6, alphabet 'all', order 6, 5 letters, solvable): lower bound 2, best certified realisation $k = 2$ via planar/none (covering group of order...
C-T-Z2p2-all	computed_rigorous	confirmed	Certified (Z2^2, alphabet 'all', order 4, 3 letters, solvable): lower bound 1, best certified realisation $k = 1$ via count/none (covering group of orde...
C-T-Z2p3-all	computed_rigorous	confirmed	Certified (Z2^3, alphabet 'all', order 8, 7 letters, solvable): lower bound 1, best certified realisation $k = 1$ via count/none (covering group of orde...
C-T-S3-transpositions	computed_rigorous	confirmed	Certified (S3, alphabet 'transpositions', order 6, 3 letters, solvable): lower bound 1, best certified realisation $k = 1$ via perm/none (covering group...
C-T-S3-all	computed_rigorous	confirmed	Certified (S3, alphabet 'all', order 6, 5 letters, solvable): lower bound 2, best certified realisation $k = 2$ via perm/none (covering group of order 6...
C-T-D4-all	computed_rigorous	confirmed	Certified (D4, alphabet 'all', order 8, 7 letters, solvable): lower bound 2, best certified realisation $k = 2$ via planar/none (covering group of order...
C-T-D5-all	computed_rigorous	confirmed	Certified (D5, alphabet 'all', order 10, 9 letters, solvable): lower bound 2, best certified realisation $k = 2$ via planar/none (covering group of orde...
C-T-Q8-all	computed_rigorous	confirmed	Certified (Q8, alphabet 'all', order 8, 7 letters, solvable): lower bound 2, best certified realisation $k = 6$ via perm/none (covering group of order 8...
C-T-A4-all	computed_rigorous	confirmed	Certified (A4, alphabet 'all', order 12, 11 letters, solvable): lower bound 2, best certified realisation $k = 2$ via perm/none (covering group of order...
C-T-S4-transpositions	computed_rigorous	confirmed	Certified (S4, alphabet 'transpositions', order 24, 6 letters, solvable): lower bound 1, best certified realisation $k = 1$ via perm/none (covering grou...
C-T-S4-all	computed_rigorous	confirmed	Certified (S4, alphabet 'all', order 24, 23 letters, solvable): lower bound 2, best certified realisation $k = 2$ via so3/none (covering group of order ...
C-T-S4-tn	computed_rigorous	confirmed	Certified (S4, alphabet 'tn', order 24, 2 letters, solvable): lower bound 2, best certified realisation $k = 2$ via so3/none (covering group of order 24...

C-T-A5-c3c5	computed_rigorous	confirmed	Certified (A5, alphabet 'c3c5', order 60, 2 letters, non-solvable): lower bound 2, best certified realisation $k = 2$ via so3/none (covering group of or...
C-T-S5-tn	computed_rigorous	confirmed	Certified (S5, alphabet 'tn', order 120, 2 letters, non-solvable): lower bound 2, best certified realisation $k = 4$ via perm/none (covering group of or...
C-T-A5-involutions	computed_rigorous	confirmed	Certified (A5, alphabet 'involutions', order 60, 15 letters, non-solvable): lower bound 1, best certified realisation $k = 1$ via so3/involutions (cover...
C-T-A5-cycles3	computed_rigorous	confirmed	Certified (A5, alphabet 'cycles3', order 60, 20 letters, non-solvable): lower bound 2, best certified realisation $k = 2$ via perm/none (covering group ...
C-T-A5-cycles5	computed_rigorous	confirmed	Certified (A5, alphabet 'cycles5', order 60, 24 letters, non-solvable): lower bound 2, best certified realisation $k = 2$ via so3/none (covering group o...
C-T-A5-all	computed_rigorous	confirmed	Certified (A5, alphabet 'all', order 60, 59 letters, non-solvable): lower bound 2, best certified realisation $k = 2$ via so3/none (covering group of or...
C-T-S5-transpositions	computed_rigorous	confirmed	Certified (S5, alphabet 'transpositions', order 120, 10 letters, non-solvable): lower bound 1, best certified realisation $k = 1$ via perm/none (coverin...
C-T-S5-all	computed_rigorous	confirmed	Certified (S5, alphabet 'all', order 120, 119 letters, non-solvable): lower bound 2, best certified realisation $k = 4$ via perm/none (covering group of...
C-T-Q8-cover	computed_rigorous	confirmed	Certified (explorer agent, re-checked by the exact verifier): $h^*(Q8, all) \leq 3$, strictly below the faithful value $h = 4$: the covering group C4:C4 of o...
C-Q8-lower	hypothesis	open	Hand argument (explorer agent, not machine-checked): $h(Q8, all) \geq 3$, hence $h(Q8, all) = 3$: lifts of i and j would be planar rotations in $SO(4)$, and...
C-atlas-improved	computed_rigorous	confirmed	17 upper bounds of the atlas were lowered by covering groups of order twice the group order (explorer agent; every certificate re-checked by the exact...
C-S5-hand3	hypothesis	open	Hand argument (explorer agent, unconditional, not machine-checked): $h(S5, all) \geq 3$ and $h(S5, tn) \geq 3$, since no finite subgroup of $SO(4)$ or $O(3)$ ma...
C-verifier	computed_rigorous	confirmed	The domain verifier passed its mandatory self-test of 46 known true and known false claims (near-boundary cases such as the same matrices without the ...
C-atlas	computed_rigorous	confirmed	Atlas over all 319 groups of order at most 63 (GAP SmallGroups library), full alphabet: own numerical character tables and GAP's exact tables give the...
C-atlas-nonsolvable	computed_rigorous	confirmed	In the atlas the only non-solvable group is A5 (order 60), with faithful $h = 2$ and $h^* = 2$ determined exactly....
C-atlas-monotone	computed_rigorous	confirmed	Non-monotonicity: 181 of the 212 solvable non-abelian groups of order at most 63 have faithful $h \geq 3$, i.e. every faithful representation of them need...
C-atlas-twist	computed_rigorous	confirmed	Sign lifts to $G \times Z_2$ (multiply a letter's matrix by -1) lower the certified upper bound below the faithful h for 37 of the 213 non-abelian groups....
C-atlas-dist	computed_rigorous	confirmed	Distribution of faithful h over the non-abelian groups of the atlas (value: number of groups): 2: 32, 3: 8, 4: 115, 5: 11, 6: 41, 8: 5, 10: 1....
C-H-protocol	observed	confirmed	Preregistered protocol (prereg.md, committed before the first confirmatory run): one-layer models diag_pos, diag_pm, hh1, hh2, hh3, hh4 (DeltaNet/Delt...
C-G-diag_pos-Z2-all	statistical	confirmed	Grid cell diag_pos on Z2/all: 0 of 20 seeds succeed at 2x-4x the longest training length (primary run on the MPS GPU; only the success count survives ...
C-G-diag_pm-Z2-all	statistical	confirmed	Grid cell diag_pm on Z2/all (mps): 18 of 20 seeds succeed at 2x-4x the longest training length (17 of 20 at 7x-8x); mean accuracy 0.969 (chance 0.500)...

C-G-hh1-Z2-all	statistical	confirmed	Grid cell hh1 on Z2/all: 18 of 20 seeds succeed at 2x-4x the longest training length (primary run on the MPS GPU; only the success count survives in t...
C-G-hh2-Z2-all	statistical	confirmed	Grid cell hh2 on Z2/all (cpu): 3 of 20 seeds succeed at 2x-4x the longest training length (9 of 20 at 7x-8x); mean accuracy 0.597 (chance 0.500); in-d...
C-G-hh3-Z2-all	statistical	confirmed	Grid cell hh3 on Z2/all (cpu): 20 of 20 seeds succeed at 2x-4x the longest training length (18 of 20 at 7x-8x); mean accuracy 0.996 (chance 0.500); in...
C-G-hh4-Z2-all	statistical	confirmed	Grid cell hh4 on Z2/all (cpu): 6 of 20 seeds succeed at 2x-4x the longest training length (7 of 20 at 7x-8x); mean accuracy 0.515 (chance 0.500); in-d...
C-G-lstm-Z2-all	statistical	confirmed	Grid cell lstm on Z2/all (cpu): 20 of 20 seeds succeed at 2x-4x the longest training length (20 of 20 at 7x-8x); mean accuracy 1.000 (chance 0.500); i...
C-G-diag_pos-Z3-all	statistical	confirmed	Grid cell diag_pos on Z3/all (mps): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.466 (chance 0.333);...
C-G-diag_pm-Z3-all	statistical	confirmed	Grid cell diag_pm on Z3/all (mps): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.604 (chance 0.333); ...
C-G-hh1-Z3-all	statistical	confirmed	Grid cell hh1 on Z3/all (mps): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.333 (chance 0.333); in-d...
C-G-hh2-Z3-all	statistical	confirmed	Grid cell hh2 on Z3/all (cpu): 10 of 20 seeds succeed at 2x-4x the longest training length (7 of 20 at 7x-8x); mean accuracy 0.519 (chance 0.333); in...
C-G-hh3-Z3-all	statistical	confirmed	Grid cell hh3 on Z3/all (cpu): 6 of 20 seeds succeed at 2x-4x the longest training length (5 of 20 at 7x-8x); mean accuracy 0.444 (chance 0.333); in-d...
C-G-hh4-Z3-all	statistical	confirmed	Grid cell hh4 on Z3/all (cpu): 12 of 20 seeds succeed at 2x-4x the longest training length (10 of 20 at 7x-8x); mean accuracy 0.698 (chance 0.333); in...
C-G-lstm-Z3-all	statistical	confirmed	Grid cell lstm on Z3/all (cpu): 19 of 20 seeds succeed at 2x-4x the longest training length (4 of 20 at 7x-8x); mean accuracy 0.950 (chance 0.333); in...
C-G-diag_pos-Z2p3-all	statistical	confirmed	Grid cell diag_pos on Z2^3/all (mps): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.126 (chance 0.125)...
C-G-diag_pm-Z2p3-all	statistical	confirmed	Grid cell diag_pm on Z2^3/all (mps): 20 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.973 (chance 0.125)...
C-G-hh1-Z2p3-all	statistical	confirmed	Grid cell hh1 on Z2^3/all (mps): 1 of 20 seeds succeed at 2x-4x the longest training length (1 of 20 at 7x-8x); mean accuracy 0.660 (chance 0.125); in...
C-G-hh2-Z2p3-all	statistical	confirmed	Grid cell hh2 on Z2^3/all (cpu): 20 of 20 seeds succeed at 2x-4x the longest training length (12 of 20 at 7x-8x); mean accuracy 0.991 (chance 0.125); ...
C-G-hh3-Z2p3-all	statistical	confirmed	Grid cell hh3 on Z2^3/all (cpu): 20 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.989 (chance 0.125); i...
C-G-hh4-Z2p3-all	statistical	confirmed	Grid cell hh4 on Z2^3/all (cpu): 20 of 20 seeds succeed at 2x-4x the longest training length (1 of 20 at 7x-8x); mean accuracy 0.981 (chance 0.125); i...
C-G-lstm-Z2p3-all	statistical	confirmed	Grid cell lstm on Z2^3/all (cpu): 20 of 20 seeds succeed at 2x-4x the longest training length (20 of 20 at 7x-8x); mean accuracy 1.000 (chance 0.125);...
C-G-diag_pos-S3-transpositions	statistical	confirmed	Grid cell diag_pos on S3/transpositions (mps): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.166 (cha...
C-G-diag_pm-S3-transpositions	statistical	confirmed	Grid cell diag_pm on S3/transpositions (mps): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.693 (chan...
C-G-hh1-S3-transpositions	statistical	confirmed	Grid cell hh1 on S3/transpositions (mps): 20 of 20 seeds succeed at 2x-4x the longest training length (18 of 20 at 7x-8x); mean accuracy 0.998 (chance...

C-G-hh2-S3-transpositions	statistical	confirmed	Grid cell hh2 on S3/transpositions (cpu): 10 of 20 seeds succeed at 2x-4x the longest training length (7 of 20 at 7x-8x); mean accuracy 0.662 (chance ...
C-G-hh3-S3-transpositions	statistical	confirmed	Grid cell hh3 on S3/transpositions (cpu): 19 of 20 seeds succeed at 2x-4x the longest training length (17 of 20 at 7x-8x); mean accuracy 0.969 (chance...
C-G-hh4-S3-transpositions	statistical	confirmed	Grid cell hh4 on S3/transpositions (cpu): 18 of 20 seeds succeed at 2x-4x the longest training length (12 of 20 at 7x-8x); mean accuracy 0.937 (chance...
C-G-lstm-S3-transpositions	statistical	confirmed	Grid cell lstm on S3/transpositions (cpu): 20 of 20 seeds succeed at 2x-4x the longest training length (20 of 20 at 7x-8x); mean accuracy 1.000 (chanc...
C-G-diag_pos-S3-all	statistical	confirmed	Grid cell diag_pos on S3/all (mps): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.166 (chance 0.167);...
C-G-diag_pm-S3-all	statistical	confirmed	Grid cell diag_pm on S3/all (mps): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.647 (chance 0.167); ...
C-G-hh1-S3-all	statistical	confirmed	Grid cell hh1 on S3/all (mps): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.303 (chance 0.167); in-d...
C-G-hh2-S3-all	statistical	confirmed	Grid cell hh2 on S3/all (cpu): 19 of 20 seeds succeed at 2x-4x the longest training length (1 of 20 at 7x-8x); mean accuracy 0.948 (chance 0.167); in-...
C-G-hh3-S3-all	statistical	confirmed	Grid cell hh3 on S3/all (cpu): 20 of 20 seeds succeed at 2x-4x the longest training length (1 of 20 at 7x-8x); mean accuracy 0.985 (chance 0.167); in-...
C-G-hh4-S3-all	statistical	confirmed	Grid cell hh4 on S3/all (cpu): 19 of 20 seeds succeed at 2x-4x the longest training length (6 of 20 at 7x-8x); mean accuracy 0.971 (chance 0.167); in-...
C-G-lstm-S3-all	statistical	confirmed	Grid cell lstm on S3/all (cpu): 20 of 20 seeds succeed at 2x-4x the longest training length (20 of 20 at 7x-8x); mean accuracy 1.000 (chance 0.167); i...
C-G-diag_pos-A5-involutions	statistical	confirmed	Grid cell diag_pos on A5/involutions (mps): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.017 (chance...
C-G-diag_pm-A5-involutions	statistical	confirmed	Grid cell diag_pm on A5/involutions (mps): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.017 (chance ...
C-G-hh1-A5-involutions	statistical	confirmed	Grid cell hh1 on A5/involutions (mps): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.017 (chance 0.01...
C-G-hh2-A5-involutions	statistical	confirmed	Grid cell hh2 on A5/involutions (cpu): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.017 (chance 0.01...
C-G-hh3-A5-involutions	statistical	confirmed	Grid cell hh3 on A5/involutions (cpu): 20 of 20 seeds succeed at 2x-4x the longest training length (3 of 20 at 7x-8x); mean accuracy 0.996 (chance 0.0...
C-G-hh4-A5-involutions	statistical	confirmed	Grid cell hh4 on A5/involutions (cpu): 20 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.990 (chance 0.0...
C-G-lstm-A5-involutions	statistical	confirmed	Grid cell lstm on A5/involutions (cpu): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.017 (chance 0.0...
C-G-diag_pos-A5-all	statistical	confirmed	Grid cell diag_pos on A5/all (mps): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.017 (chance 0.017);...
C-G-diag_pm-A5-all	statistical	confirmed	Grid cell diag_pm on A5/all (mps): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.017 (chance 0.017); ...
C-G-hh1-A5-all	statistical	confirmed	Grid cell hh1 on A5/all (mps): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.017 (chance 0.017); in-d...
C-G-hh2-A5-all	statistical	confirmed	Grid cell hh2 on A5/all (cpu): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.017 (chance 0.017); in-d...

C-G-hh3-A5-all	statistical	confirmed	Grid cell hh3 on A5/all (cpu): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.017 (chance 0.017); in-d...
C-G-hh4-A5-all	statistical	confirmed	Grid cell hh4 on A5/all (cpu): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.017 (chance 0.017); in-d...
C-G-lstm-A5-all	statistical	confirmed	Grid cell lstm on A5/all (cpu): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.017 (chance 0.017); in-...
C-G-diag_pos-S5-transpositions	statistical	confirmed	Grid cell diag_pos on S5/transpositions (mps): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.008 (cha...
C-G-diag_pm-S5-transpositions	statistical	confirmed	Grid cell diag_pm on S5/transpositions (mps): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.016 (chan...
C-G-hh1-S5-transpositions	statistical	confirmed	Grid cell hh1 on S5/transpositions (mps): 20 of 20 seeds succeed at 2x-4x the longest training length (20 of 20 at 7x-8x); mean accuracy 0.999 (chance...
C-G-hh2-S5-transpositions	statistical	confirmed	Grid cell hh2 on S5/transpositions (cpu): 19 of 20 seeds succeed at 2x-4x the longest training length (17 of 20 at 7x-8x); mean accuracy 0.955 (chance...
C-G-hh3-S5-transpositions	statistical	confirmed	Grid cell hh3 on S5/transpositions (cpu): 20 of 20 seeds succeed at 2x-4x the longest training length (14 of 20 at 7x-8x); mean accuracy 1.000 (chance...
C-G-hh4-S5-transpositions	statistical	confirmed	Grid cell hh4 on S5/transpositions (cpu): 19 of 20 seeds succeed at 2x-4x the longest training length (11 of 20 at 7x-8x); mean accuracy 0.964 (chance...
C-G-lstm-S5-transpositions	statistical	confirmed	Grid cell lstm on S5/transpositions (cpu): 8 of 20 seeds succeed at 2x-4x the longest training length (1 of 20 at 7x-8x); mean accuracy 0.853 (chance ...
C-G-diag_pos-S5-all	statistical	confirmed	Grid cell diag_pos on S5/all (mps): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.008 (chance 0.008);...
C-G-diag_pm-S5-all	statistical	confirmed	Grid cell diag_pm on S5/all (mps): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.008 (chance 0.008); ...
C-G-hh1-S5-all	statistical	confirmed	Grid cell hh1 on S5/all (mps): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.008 (chance 0.008); in-d...
C-G-hh2-S5-all	statistical	confirmed	Grid cell hh2 on S5/all (cpu): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.008 (chance 0.008); in-d...
C-G-hh3-S5-all	statistical	confirmed	Grid cell hh3 on S5/all (cpu): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.008 (chance 0.008); in-d...
C-G-hh4-S5-all	statistical	confirmed	Grid cell hh4 on S5/all (cpu): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.008 (chance 0.008); in-d...
C-G-lstm-S5-all	statistical	confirmed	Grid cell lstm on S5/all (cpu): 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.008 (chance 0.008); in-...
C-H-accuracy	statistical	confirmed	Agreement of each predictor with the cell outcomes on the determined cells: our algebraic predictor 47 of 61; circuit class (solvable) 41 of 61; abeli...
C-H-D1	statistical	confirmed	Preregistered test D1 (one-sided Fisher exact test): cell ['hh1', 'A5', 'involutions'] vs control ['hh1', 'A5', 'all'], successful seeds [0, 0], p = 1...
C-H-D2	statistical	confirmed	Preregistered test D2 (one-sided Fisher exact test): cell ['hh2', 'A5', 'all'] vs control ['hh1', 'A5', 'all'], successful seeds [0, 0], p = 1.000, Be...
C-H-D3	statistical	confirmed	Preregistered test D3 (one-sided Fisher exact test): cell ['hh1', 'Z2^3', 'all'] vs control ['hh1', 'Z3', 'all'], successful seeds [1, 0], p = 0.500, ...
C-H-D4	statistical	confirmed	Preregistered test D4 (one-sided Fisher exact test): cell ['hh1', 'S5', 'transpositions'] vs control ['hh1', 'S5', 'all'], successful seeds [20, 0], p...

C-H-D5	statistical	confirmed	Preregistered test D5 (one-sided Fisher exact test): cell ['hh1', 'Z2', 'all'] vs control ['hh1', 'Z3', 'all'], successful seeds [18, 0], $p = 1.7e-09$...
C-H-D6	statistical	confirmed	Preregistered test D6 (one-sided Fisher exact test): cell ['diag_pm', 'Z2^3', 'all'] vs control ['diag_pm', 'Z3', 'all'], successful seeds [20, 0], $p = 1.000$, Benjamini-Hochberg adjust...
C-H-NC-hh4-S5	statistical	confirmed	Preregistered test NC-hh4-S5 (negative control, random targets): cell ['hh4', 'S5', 'all'], successful seeds [0], $p = 1.000$, Benjamini-Hochberg adjust...
C-H-NC-lstm-S5	statistical	confirmed	Preregistered test NC-lstm-S5 (negative control, random targets): cell ['lstm', 'S5', 'all'], successful seeds [0], $p = 1.000$, Benjamini-Hochberg adjust...
C-H-NC-hh2-A5	statistical	confirmed	Preregistered test NC-hh2-A5 (negative control, random targets): cell ['hh2', 'A5', 'all'], successful seeds [0], $p = 1.000$, Benjamini-Hochberg adjust...
C-X-protocol	observed	confirmed	Preregistered addendum H-EX2 (committed before its runs, prereg.md 2026-10-04 05:00): the closest prior work's generator formats S4/tn (a transpositio...
C-X-hh1-A5-c3c5	statistical	confirmed	H-EX2 cell hh1 on A5/c3c5: 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.017 (chance 0.017); in-distr...
C-X-hh2-A5-c3c5	statistical	confirmed	H-EX2 cell hh2 on A5/c3c5: 18 of 20 seeds succeed at 2x-4x the longest training length (18 of 20 at 7x-8x); mean accuracy 0.901 (chance 0.017); in-dis...
C-X-hh3-A5-c3c5	statistical	confirmed	H-EX2 cell hh3 on A5/c3c5: 18 of 20 seeds succeed at 2x-4x the longest training length (10 of 20 at 7x-8x); mean accuracy 0.923 (chance 0.017); in-dis...
C-X-hh1-S4-tn	statistical	confirmed	H-EX2 cell hh1 on S4/tn: 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.225 (chance 0.042); in-distrib...
C-X-hh2-S4-tn	statistical	confirmed	H-EX2 cell hh2 on S4/tn: 6 of 20 seeds succeed at 2x-4x the longest training length (4 of 20 at 7x-8x); mean accuracy 0.347 (chance 0.042); in-distrib...
C-X-hh3-S4-tn	statistical	confirmed	H-EX2 cell hh3 on S4/tn: 19 of 20 seeds succeed at 2x-4x the longest training length (3 of 20 at 7x-8x); mean accuracy 0.945 (chance 0.042); in-distri...
C-X-E1	statistical	confirmed	H-EX2 test E1 (one-sided Fisher exact): ['hh2', 'A5', 'c3c5'] vs ['hh1', 'A5', 'c3c5'], successful seeds [18, 0], $p = 1.7e-09$, BH-adjusted $p = 2.5e-09$...
C-X-E2	statistical	confirmed	H-EX2 test E2 (one-sided Fisher exact): ['hh2', 'S4', 'tn'] vs ['hh1', 'S4', 'tn'], successful seeds [6, 0], $p = 0.010$, BH-adjusted $p = 0.010$ ($m = 3$, ...)
C-X-E3	statistical	confirmed	H-EX2 test E3 (one-sided Fisher exact): ['hh3', 'A5', 'c3c5'] vs ['hh1', 'A5', 'c3c5'], successful seeds [18, 0], $p = 1.7e-09$, BH-adjusted $p = 2.5e-09$...
C-X-result	statistical	confirmed	H-EX2 outcome: our predictor matches 5 of 6 cells, the permutation-representation law matches 4 of 6; preregistered success criterion of H-EX2: False...
C-H-exploratory	observed	confirmed	Exploratory analysis (post hoc, not preregistered): restricted to the 5 tasks on which the LSTM positive control succeeded (S3/all, S3/transpositions,...
C-H-replication	statistical	confirmed	Device replication (the same protocol and seeds run twice, once on the MPS GPU and once on the CPU, because of a runner bug, prereg addendum 07:00): d...
C-H-gates	statistical	confirmed	Preregistered success criteria: accuracy beats every baseline: False; all six discriminating tests significant after BH: False; negative control clean...
C-expressivity-R2	computed_rigorous	confirmed	Lab round 2. Question: For the abelian group $Z2^3$ with alphabet 'all', compare $h(Z2^3, \text{all})$ (faithful representations only) with $h^*(Z2^3, \text{all})$. Which ...
C-expressivity-R2-I	hypothesis	open	Uninspected interpretation by the agent for expressivity-R2 (not a result; NOT consistent with the hardened consistency rule): $h(Z2^3, \text{all}) = 3$ (faith...
C-expressivity-R3	computed_rigorous	confirmed	Lab round 3. Question: Is $h^*(A5, \text{involutions})$ at 1 or above if only a single Householder product per token (HH_1) is allowed, and does the answer stil...

C-expressivity-R3-I	hypothesis	open	Uninspected interpretation by the agent for expressivity-R3 (not a result; NOT consistent with the hardened consistency rule): > 1 ...
C-expressivity-R4	computed_rigorous	contested by red-team counter-check	Lab round 4. Question: For the quaternion group Q_8 with alphabet 'all', what is $h(Q_8, \text{all})$, and how far apart are the certified bounds on $h^*(Q_8, \text{all})$?...
C-expressivity-R4-I	hypothesis	open	Uninspected interpretation by the agent for expressivity-R4 (not a result; NOT consistent with the hardened consistency rule): $h(Q_8, \text{all}) = 4$...
C-expressivity-R5	computed_rigorous	confirmed	Lab round 5. Question: Can a single layer with 2 Householder reflections per token track the non-solvable group A_5 with alphabet 'all' exactly? Determ...
C-expressivity-R5-I	hypothesis	open	Uninspected interpretation by the agent for expressivity-R5 (not a result; consistent with the hardened consistency rule): Yes, 2 Householder reflecti...
C-expressivity-R6	computed_rigorous	contested by red-team counter-check	Lab round 6. Question: How does $h(A_5, \text{involutions})$ compare with $h(A_5, \text{all})$ when only the involutions are allowed as the alphabet, and is the value 1...
C-expressivity-R6-I	hypothesis	open	Uninspected interpretation by the agent for expressivity-R6 (not a result; NOT consistent with the hardened consistency rule): 2...
C-expressivity-R7	computed_rigorous	confirmed	Lab round 7. Question: For S_4 with alphabet 'all', the permutation representation needs $\text{rank}(P(s) - I) = 3$ for 4-cycles. Is the true $h^*(S_4, \text{all})$ small...
C-expressivity-R7-I	hypothesis	open	Uninspected interpretation by the agent for expressivity-R7 (not a result; consistent with the hardened consistency rule): ja, $h^*(S_4, \text{all}) = 2$...
C-expressivity-R8	computed_rigorous	contested by red-team counter-check	Lab round 8. Question: Which groups with involutions as alphabet (e.g. S_4 , D_4 , Z_2^2 , or Q_8 with a suitable twist) have $h^*(G, \text{involutions}) = 1$, and can...
C-expressivity-R8-I	hypothesis	open	Uninspected interpretation by the agent for expressivity-R8 (not a result; consistent with the hardened consistency rule): Z_2^2 ...
C-expressivity-R9	computed_rigorous	confirmed	Lab round 9. Question: For S_5 with alphabet 'all', what are the best certified lower and upper bounds on $h^*(S_5, \text{all})$, and what is $h(S_5, \text{all})$ over fait...
C-expressivity-R9-I	hypothesis	open	Uninspected interpretation by the agent for expressivity-R9 (not a result; NOT consistent with the hardened consistency rule): $h^*(S_5, \text{all})$: lower bound...
C-expressivity-R15	computed_rigorous	confirmed	Lab round 15. Question: How does $h^*(A_5, \text{all}) = 2$ compare with $h(A_5, \text{involutions}) = 2$ when the involution alphabets are compared with the completeness ...
C-expressivity-R15-I	hypothesis	open	Uninspected interpretation by the agent for expressivity-R15 (not a result; consistent with the hardened consistency rule): No distance detected ($h^*(A...$
C-neg1	observed	confirmed	Negative result of the lab (no claim passed the verifier and the consistency rule): [F3] Is one Householder reflection per token (plain DeltaNet with ...
C-neg2	observed	confirmed	Negative result of the lab (no claim passed the verifier and the consistency rule): [F25] Does the bound $h^*(S_5, \text{all}) \leq 4$ also hold without a permutati...
C-neg3	observed	confirmed	Negative result of the lab (no claim passed the verifier and the consistency rule): [F26] How does $h(S_4, \text{all})$ and $h^*(S_4, \text{all})$ compare with S_5 , and can...
C-neg4	observed	confirmed	Negative result of the lab (no claim passed the verifier and the consistency rule): [F13] Does $h^*(A_5, \text{involutions}) = 2$ still hold when, instead of fai...
C-neg5	observed	confirmed	Negative result of the lab (no claim passed the verifier and the consistency rule): [F21] Does $h^*(S_4, \text{all}) = 2$ still hold even when only involutions a...
C-neg6	observed	confirmed	Negative result of the lab (no claim passed the verifier and the consistency rule): [F11] Does $h(Z_2^3, \text{all}) = 3$ still hold even when the range of fai...
C-lab	observed	confirmed	The agentic lab (literature scout with code-checked quotes, integrator, cascade of four researcher agents, exact verifier, red-team agent, per-round p...

C-loophole1	observed	confirmed	Loophole found in lab round 1 and closed: an agent answered ‘no’ (one reflection is not enough for A5 with involution inputs) and backed it with two t...
C-redteam	observed	confirmed	An independent red-team agent (read and execute only) found three bugs: group names S1, D1, D2 denoted wrong groups (one false sentence each could be ...
C-lit-merrill-tc0	observed	confirmed	Reference (arXiv:2207.00729, title verified by code against arXiv): “The Parallelism Tradeoff: Limitations of Log-Precision Transformers”. Verbatim ab...
C-lit-illusion	observed	confirmed	Reference (arXiv:2404.08819, title verified by code against arXiv): “The Illusion of State in State-Space Models”. Verbatim abstract quote checked by ...
C-lit-grazzi	observed	confirmed	Reference (arXiv:2411.12537, title verified by code against arXiv): “Unlocking State-Tracking in Linear RNNs Through Negative Eigenvalues”. Verbatim a...
C-lit-deltaproduct	observed	confirmed	Reference (arXiv:2502.10297, title verified by code against arXiv): “DeltaProduct: Improving State-Tracking in Linear RNNs via Householder Products”. ...
C-lit-rwkv7	observed	confirmed	Reference (arXiv:2503.14456, title verified by code against arXiv): “RWKV-7”Goose” with Expressive Dynamic State Evolution”. Verbatim abstract quote ...
C-lit-sarrof	observed	confirmed	Reference (arXiv:2405.17394, title verified by code against arXiv): “The Expressive Capacity of State Space Models: A Formal Language Perspective”. Ve...
C-lit-shakerinava	observed	confirmed	Reference (arXiv:2603.01959, title verified by code against arXiv): “The Expressive Limits of Diagonal SSMs for State-Tracking”. Verbatim abstract quo...
C-lit-howe	observed	confirmed	Reference (arXiv:2609.18966, title verified by code against arXiv): “The Automaton Underneath: The Additive Input Pathway Is a Parasitic Attractor for...
C-lit-liu	observed	confirmed	Reference (arXiv:2210.10749, title verified by code against arXiv): “Transformers Learn Shortcuts to Automata”. Verbatim abstract quote checked by cod...
C-lit-cot	observed	confirmed	Reference (arXiv:2310.07923, title verified by code against arXiv): “The Expressive Power of Transformers with Chain of Thought”. Verbatim abstract qu...
C-lit-hahn	observed	confirmed	Reference (arXiv:1906.06755, title verified by code against arXiv): “Theoretical Limitations of Self-Attention in Neural Sequence Models”. Verbatim ab...
C-lit-deletang	observed	confirmed	Reference (arXiv:2207.02098, title verified by code against arXiv): “Neural Networks and the Chomsky Hierarchy”. Verbatim abstract quote checked by co...
C-lit-li-cot	observed	confirmed	Reference (arXiv:2402.12875, title verified by code against arXiv): “Chain of Thought Empowers Transformers to Solve Inherently Serial Problems”. Verb...
C-lit-deltaproduct-body	observed	confirmed	Reference arXiv:2502.10297 (full-text quote checked by code against the arXiv HTML): “Unexpectedly, S 4 and A 5 can extrapolate robustly using only n ...
C-lit-deltaproduct-so3-body	observed	confirmed	Reference arXiv:2502.10297 (full-text quote checked by code against the arXiv HTML): “This efficiency arises from their isomorphism to subgroups of SO...
C-lit-howe-law-body	observed	confirmed	Reference arXiv:2609.18966 (full-text quote checked by code against the arXiv HTML): “the minimal n h that length-generalizes equals the maximal gener...
C-lit-howe-format-body	observed	confirmed	Reference arXiv:2609.18966 (full-text quote checked by code against the arXiv HTML): “S 4 (transposition + 4-cycle), A 5 (3-cycle + 5-cycle; all-even)...

C-lit-rwkv7-swaps-body	observed	confirmed	Reference arXiv:2503.14456 (full-text quote checked by code against the arXiv HTML): “RWKV-7 can, by Lemma 2 , solve the problem of tracking swaps on ...
C-lit-grazzi-cd-body	observed	confirmed	Reference arXiv:2411.12537 (full-text quote checked by code against the arXiv HTML): “every $n \times n$ orthogonal matrix can be written as a product of n r...
C-lit-ckda-thm4-body	observed	confirmed	Reference arXiv:2609.24797 (full-text quote checked by code against the arXiv HTML): “We rule out one-layer S 5 tracking for CKDA and DeltaProduct k (...
C-lit-ckda-four-body	observed	confirmed	Reference arXiv:2609.24797 (full-text quote checked by code against the arXiv HTML): “this proves that four is the minimum number of Householder trans...
C-lit-ckda-compression-body	observed	confirmed	Reference arXiv:2609.24797 (full-text quote checked by code against the arXiv HTML): “We first choose a word whose linear part has minimum Frobenius n...
C-lit-ckda-decoder-body	observed	confirmed	Reference arXiv:2609.24797 (full-text quote checked by code against the arXiv HTML): “Tracking does not require a faithful representation: a many-to-o...
C-lit-ckda-so3-body	observed	confirmed	Reference arXiv:2609.24797 (full-text quote checked by code against the arXiv HTML): “A single CKDA layer (one head) tracks every finite group isomorp...
C-lit-martino-singh-body	observed	confirmed	Reference arXiv:1804.05089 (full-text quote checked by code against the arXiv HTML): “We say that a group G is generated (resp. strictly generated) ...
C-lit-lange-mikhailova-body	observed	confirmed	Reference arXiv:1509.06922 (full-text quote checked by code against the arXiv HTML): “We survey the existing parts of a classification of finite group...
C-lit-barrington	observed	confirmed	Reference (doi:10.1016/0022-0000(89)90037-8, title verified by code against Crossref): “Bounded-width polynomial-size branching programs recognize exa...